

B.M.S COLLEGE OF ENGINEERING BENGALURU

Autonomous Institute, Affiliated to VTU



IEE AAT

REPORT ON: TYPES OF ELECTRICAL WIRES

Submitted in partial fulfillment of the requirements for AAT

**Bachelor of Engineering
in
Computer Science and Engineering**

Submitted by:

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2024-2025

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DEPARTMENT OF ELECTRICAL AND ELECTRONICS



DECLARATION

I, **Ayush kumar Prajapati** students of 1st Semester, B.E, Department of CSBS, BMS College of Engineering, Bangalore, hereby declare that, this AAT Project entitled "TYPES OF ELECTRICAL WIRE" has been carried out in Department of EEE, BMS College of Engineering, Bangalore during the academic semester Oct 2024 – Jan 2025. I also declare that to the best of my knowledge and belief, the AAT Project report is not from part of any other report by any other students.

Student Name

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CERTIFICATE

This is to certify that the AAT Project titled “TYPES OF ELECTRICAL WIRE” has been carried out by **Ayush** during the academic year 2024-2025.

Signature of the Faculty in Charge:

Table of Contents:

S.NO	Content	Page No
1	Introduction	1
2	Types of wires	1
	2.1 Vulcanized Indian Rubber (V.I.R.) Wires	1
	2.2 Tough Rubber Sheathed (T.R.S.) Wires	2
	2.3 Poly Vinyl Chloride (P.V.C.) Wires	2
	2.4 Lead Alloy Sheathed Wires	3
	2.5 Weatherproof Wires	3
	2.6 Flexible Wires	4
3	Summary Table	5
4	Conclusion	6

REPORT ON TYPES OF WIRES

1. Introduction

Wires are an essential part of electrical systems, providing a medium for current flow in domestic, commercial, and industrial applications. The type of wire used in an electrical installation depends on factors such as voltage rating, insulation, environmental conditions, and mechanical strength.

This report discusses the different types of wires mentioned in the **Basic Electrical Engineering (BEE) document**, including **Vulcanized Indian Rubber (V.I.R.) Wires, Tough Rubber Sheathed (T.R.S.) Wires, Poly Vinyl Chloride (P.V.C.) Wires, Lead Alloy Sheathed Wires, Weatherproof Wires, and Flexible Wires**. Each type has specific characteristics, advantages, and applications, which are described in the following sections.

2. Types of Wires

2.1 Vulcanized Indian Rubber (V.I.R.) Wires

Construction and Features

- V.I.R. wires consist of conductors insulated with **vulcanized rubber**.
- The conductor is **tinned** to prevent the rubber from sticking to it.
- The thickness of the rubber insulation varies based on the operating voltage.
- A **cotton braiding** is applied over the rubber for added **moisture protection**.
- These wires are finished with wax for cleanliness.

Applications

- Used in **low and medium voltage** applications.
- Found in **older electrical installations** before the introduction of modern insulated wires.

Advantages

- Provides good electrical **insulation**.
- Offers **moderate flexibility**, making it easy to install.

Disadvantages

- **Deteriorates over time**, especially in **high-temperature environments**.
- **Not suitable** for modern electrical appliances.
- Has been replaced by **PVC-insulated wires** due to better performance and cost-effectiveness.

2.2 Tough Rubber Sheathed (T.R.S.) Wires

Construction and Features

- An improved version of **V.I.R. wires** with an **additional sheath of tough rubber**.
- The tough rubber provides **extra protection** against **moisture, wear, and mechanical damage**.
- Available in **single-core, twin-core, and three-core** versions.

Applications

- Used in **industrial and commercial** electrical installations.
- Common in **temporary outdoor wiring**, such as **construction sites**.

Advantages

- Offers **better mechanical protection** compared to **V.I.R. wires**.
- More **durable** and provides **better insulation**.

Disadvantages

- **Not resistant to high temperatures** and can degrade over time.
- **Has been replaced** by **PVC wires** in modern applications.

2.3 Poly Vinyl Chloride (P.V.C.) Wires

Construction and Features

- The most commonly used wires today.
- Insulated with **Poly Vinyl Chloride (PVC)**, a **thermoplastic material**.
- **Resistant to moisture, chemicals, and mechanical wear**.

Applications

- **Household wiring** and **industrial electrical systems**.
- Suitable for **both indoor and outdoor** use.

Advantages

- **Durable and moisture-resistant**.
- **Cost-effective** compared to rubber-insulated wires.
- Available in **flame-retardant variants** for **fire safety**.

Disadvantages

- **Softens at high temperatures**, limiting its use in heat-intensive environments.
 - Can release **toxic fumes when burned**.
-

2.4 Lead Alloy Sheathed Wires

Construction and Features

- These wires have an **outer lead sheath** for protection.
- The sheath is very thin (approximately **0.12 cm thick**).
- Used in **moisture-prone environments** where rubber or PVC insulation might fail.

Applications

- Commonly used in **underground installations**.
- Suitable for **industries and damp environments** where ordinary insulation is ineffective.

Advantages

- **Highly resistant to moisture and chemical exposure**.
- Provides **mechanical protection**.

Disadvantages

- **Heavy and less flexible** compared to other wires.
 - **Expensive**, making it less commonly used today.
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2.5 Weatherproof Wires

Construction and Features

- Specifically designed for **outdoor use**.
- The conductor is **not tinned** but covered with **three layers of fibrous yarn** saturated with a **waterproof compound**.
- Provides **good resistance** against **rain, humidity, and UV radiation**.

Applications

- Used for **service connections** from **overhead power lines** to buildings.
- Suitable for **agricultural applications** and **outdoor lighting**.

Advantages

- **Highly durable** for outdoor use.
- Resistant to **weather conditions**, including UV exposure.

Disadvantages

- **More expensive** than standard PVC wires.
 - **Requires careful handling** to avoid damage to the insulation.
-

2.6 Flexible Wires

Construction and Features

- Made up of **multiple fine strands** instead of a **single solid conductor**.
- Insulated with **PVC** for durability and flexibility.
- Commonly used in **applications requiring frequent movement**.

Applications

- **Household electrical appliances** like **fans, irons, and refrigerators**.
- Used in **automotive wiring** and **portable electrical devices**.

Advantages

- **Highly flexible**, making it easy to install in tight spaces.
- **Durable and resistant** to mechanical stress.

Disadvantages

- **Not suitable for high-voltage applications**.
 - **More expensive** compared to **solid-core wires**.
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3. Summary Table

Type of Wire	Insulation Material	Main Features	Common Uses
V.I.R. Wires	Vulcanized Rubber	Flexible, moisture-resistant	Older domestic wiring
T.R.S. Wires	Tough Rubber	Extra insulation, durable	Industrial wiring, temporary outdoor installations
P.V.C. Wires	Polyvinyl Chloride	Most commonly used, fire-resistant	Household, commercial, and industrial wiring
Lead Alloy Sheathed Wires	Lead	Moisture-resistant, used in damp environments	Underground wiring, industrial applications
Weatherproof Wires	Fibrous Yarn + Waterproof Compound	Outdoor use, UV-resistant	Overhead power connections, agricultural wiring
Flexible Wires	PVC	High flexibility, used in moving appliances	Portable electrical appliances, automotive wiring

4. Conclusion

The selection of electrical wires depends on several factors, including **insulation material, mechanical strength, environmental conditions, and application requirements**.

- **PVC wires** are the most widely used due to their **durability, affordability, and safety**.
- **Weatherproof and lead-sheathed wires** are chosen for **specialized applications** requiring **high resistance to moisture**.
- **Flexible wires** are preferred for **appliances and automotive use** due to their **high flexibility**.

Understanding these types of wires and their properties ensures **safe and efficient electrical installations**. Proper selection, installation, and maintenance of wires help **prevent electrical hazards** and **enhance the lifespan of electrical systems**.